

Esteban Sitek

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with **12+** years building and operating production data systems across AdTech, e-commerce, and healthcare. Expert in high-volume event pipelines, **Kafka**, **ClickHouse**, and distributed systems architecture. Proven track record leading data infrastructure modernization, mentoring engineers, and delivering near-real-time analytics platforms serving millions of users. Ready to architect scalable, observable data solutions and drive technical excellence.

SKILLS

- Languages:** Python, SQL, Java, JavaScript, TypeScript, Bash, HTML, CSS
- Data & ML:** Apache Kafka, ClickHouse, Apache Airflow, Apache Spark, Pandas, dbt, NumPy, Elasticsearch
- Backend:** FastAPI, Node.js, Express, Django, Spring Boot, NestJS
- Database:** PostgreSQL, MySQL, MongoDB, Redis, Cassandra, DynamoDB
- Testing:** Pytest, Jest, Cypress, Selenium, k6, JMeter
- Cloud & DevOps:** AWS, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, AWS Lambda, S3, EC2, RDS, CloudWatch, Linux
- Others:** RESTful API, GraphQL, Git, GitHub, Jira, Confluence, Datadog, Prometheus, Grafana, Apache Superset

EXPERIENCE

Publicis Sapient

Senior Software Engineer

Jun 2024–Present

- Led the architectural redesign of a legacy batch processing system, introducing **Kafka** topics for event streaming and **Python**-based consumer workers that reduced end-to-end latency from 6 hours to under 15 minutes for high-volume transactional data.
- Designed and deployed an **Apache Airflow** orchestration layer managing 40+ interdependent ETL workflows across financial services clients, implementing dynamic DAG generation to handle schema variations and enabling rollback capabilities that prevented data loss during failed jobs.
- Mentored three junior data engineers on distributed systems concepts, **SQL** query optimization, and production debugging practices, establishing code review standards that reduced production incidents by **32%** and improved average pull request resolution time to under 24 hours.
- Built real-time aggregation pipelines using **ClickHouse** that ingested 500k+ events per minute, optimizing table schemas with appropriate partitioning strategies and leveraging ReplacingMergeTree engines to handle late-arriving corrections while maintaining sub-second query performance for analytics dashboards.

- Implemented comprehensive data quality monitoring using **dbt** tests and **Prometheus** metrics, creating alerts for anomalies in fact table row counts, dimension cardinality, and join key distributions that caught a revenue attribution bug affecting \$2M+ of quarterly reporting before stakeholder visibility.
- Collaborated with product and analytics teams to translate business requirements into scalable data models, owning the full lifecycle from **SQL** schema design through **Grafana** dashboard rollout, reducing analytics team query turnaround from 2 days to 2 hours.
- Refactored a homegrown CDC pipeline to use **Kafka** Connect, eliminating the need for custom polling logic and improving data freshness from hourly snapshots to sub-minute change capture across **PostgreSQL** and **MySQL** source systems.
- Defined and operationalized SLOs for data pipelines including end-to-end latency, freshness, and completeness, implementing monitoring stacks with **Datadog** and **CloudWatch** that gave leadership visibility into system health and enabled proactive capacity planning.
- Drove infrastructure-as-code adoption by migrating manual **AWS** resource provisioning to **Terraform**, reducing environment setup time from 2 weeks to 2 hours and enabling self-service data sandbox creation for analytics experiments.
- Owned the incident response workflow for production data issues, establishing a runbook system and pair-debugging sessions that reduced MTTR for high-impact data discrepancies by **18%** and improved post-incident review documentation.
- Extended **Kubernetes** cluster observability by adding sidecar logging to data job containers, shipping structured logs to **Elasticsearch** and **Grafana**, which made debugging production job failures significantly faster during high-traffic periods.
- Investigated and resolved a critical issue in the event replay mechanism where **Kafka** offset commits were not idempotent, rebuilding consumer restart logic with explicit state management that prevented duplicate aggregations and data inconsistencies.

CareSimple

Oct 2022–May 2024

Senior Full Stack Engineer

- Spearheaded the build of an analytics platform processing patient monitoring and clinical outcome data using **Kafka** event streams feeding into a **ClickHouse** OLAP cluster, enabling real-time dashboards for 50+ healthcare providers tracking population health metrics and readmission trends.
- Owned the data modeling layer in **ClickHouse**, designing efficient schemas with SummingMergeTree and AggregatingMergeTree tables for pre-computed patient cohorts and clinical metrics, achieving **40%** reduction in query response times for compliance reporting workloads.
- Developed and maintained **Apache Airflow** DAGs orchestrating daily ETL jobs that merged structured clinical data from **PostgreSQL** with unstructured monitoring records, implementing backfill logic that recovered 6 months of lost data without disrupting production schedules.
- Reduced patient data export latency from 30 minutes to under 2 minutes by designing a **Redis**-backed caching layer for frequently requested cohort queries in **ClickHouse**, dramatically improving the analytics team's ability to respond to urgent provider requests.

- Built validation and reconciliation workflows using **Python** and **Pandas** that compared clinical metric calculations against source system truth, automatically detecting data drift and alerting engineers to investigate upstream issues before patient-facing reports were published.
- Coordinated with healthcare compliance and security teams to implement audit logging for all analytics pipeline stages using structured **JSON** events sent to **Kafka**, enabling full traceability of which data was accessed, transformed, and presented to external stakeholders.
- Improved pipeline reliability by adding comprehensive error handling and retry strategies across 25+ **Node.js** backend services that fed data into the analytics platform, reducing silent failures and making error propagation visible through **Datadog** monitoring.
- Partnered with the analytics team to migrate legacy **MySQL** reporting queries to optimized **ClickHouse** equivalents, combining schema redesign with query rewriting that eliminated timeouts affecting monthly executive reporting and reduced database CPU load by **25%**.
- Established data freshness expectations and monitoring for the analytics platform, implementing **Prometheus** metrics tracking pipeline stage latencies and alerting when any stage exceeded SLA thresholds, improving stakeholder trust in near-real-time health metrics.
- Designed a disaster recovery procedure for the **ClickHouse** cluster that included regular backups to **S3**, tested recovery procedures monthly, and documented runbooks that enabled successful full cluster recovery in under 4 hours during a production failure.

Shopee

Jul 2018–Sep 2022

Full Stack Engineer

- Contributed to the analytics infrastructure team building ETL pipelines that ingested hundreds of millions of daily e-commerce events into a **ClickHouse** cluster serving dashboards for pricing, seller performance, and conversion analysis across Southeast Asian markets.
- Implemented **Kafka** producers and consumer groups in **Python** that reliably delivered user behavior events (searches, clicks, purchases) from frontend and mobile applications into the data lake, handling at-least-once delivery guarantees and implementing idempotent processing logic.
- Built **Apache Spark** jobs that aggregated daily transaction events into dimensional fact tables in **PostgreSQL**, applying window functions and event deduplication logic to produce accurate seller commission calculations and revenue attribution reports.
- Optimized **ClickHouse** queries supporting the seller analytics dashboard by redesigning table partitioning schemes and implementing materialized views for common aggregations, cutting query latency from 8 seconds to under 400ms and enabling real-time metric exploration.
- Developed a data quality framework using **Pandas** and **SQL** that detected anomalies in daily event volumes, attribution metrics, and conversion funnels, surfacing issues to the data engineering team within minutes of observation rather than hours of manual investigation.
- Maintained and iterated on data pipelines written in **Airflow** that processed 50+ terabytes of historical event data during monthly reconciliations, implementing backfill logic and reprocessing capabilities that recovered consistency after upstream schema changes.

- Collaborated with fraud detection and risk teams to add real-time event enrichment logic in **Kafka** consumers, flagging suspicious transaction patterns immediately after purchase events were captured, reducing fraud-related chargebacks by **12%** in the travel category.
- Designed a **Redis** caching layer for frequently accessed seller metrics in **ClickHouse**, dramatically improving analytics dashboard responsiveness during peak traffic periods when 10k+ merchants were simultaneously checking real-time sales and conversion metrics.
- Contributed to backend services using **Node.js** and **Express** that exposed analytics data through **RESTful API** endpoints, implementing aggregation logic, filtering, and sorting to serve mobile and web clients while maintaining **MongoDB** as the operational store for low-latency lookups.
- Participated in on-call rotation for the analytics platform, triaging **ClickHouse** query timeouts, **Kafka** consumer lag issues, and data freshness alerts, which built deep understanding of distributed systems failure modes and recovery strategies.

Altitude Games

Nov 2014–Jul 2018

Full Stack Engineer

- Built event analytics pipelines in **Python** that ingested millions of daily user behavior events (game starts, level completions, purchases, churn) from multiplayer game clients into a data warehouse, enabling the analytics team to track engagement metrics and monetization trends across titles.
- Designed and maintained **MySQL** schemas for game state and user activity, implementing indexes and query optimization that supported real-time leaderboard calculations and cohort analysis serving 500k+ daily active players across multiple continents.
- Developed **Node.js** backend services handling event streaming and aggregation, integrating **Kafka** message queues for reliable event delivery from game clients and implementing consumer workers that transformed raw events into user session records stored in **Redis** and **PostgreSQL**.
- Implemented data quality checks using **SQL** queries and custom **Python** scripts that detected data corruption, missing events, and timestamp anomalies in the game telemetry pipeline, enabling early identification of client bugs before they cascaded into analytics issues.
- Contributed to infrastructure automation using **Bash** scripts and basic **Terraform** templates that provisioned **AWS** resources including **EC2** instances for game servers, **S3** buckets for backup storage, and **RDS** instances for the operational database, reducing environment setup effort.
- Collaborated with the gameplay and monetization teams to design the event schema capturing user actions, item purchases, and progression milestones, ensuring events contained sufficient context for downstream attribution and cohort analysis without bloating message sizes.

EDUCATION

Ateneo de Manila University

Aug 2010–Jul 2014

Bachelor of Science in Computer Science